In the above program class "result " is derived from class "exam" which itself is derived from class student.

Note

In multilevel inheritance the level of inheritance can be extended to any number of level depending upon the relation. Multilevel inheritance is similar to relation between grandfather, father and child.

A class without any declaration will have 1 byte size.class x{}; X occupies 1 byte.

In the above program the derived class "result" has acquired the properties of class "detail" and class "exam" which is derived from "student". So this inheritance is a combination of multi level and multiple inheritance and so it is called hybrid inheritance

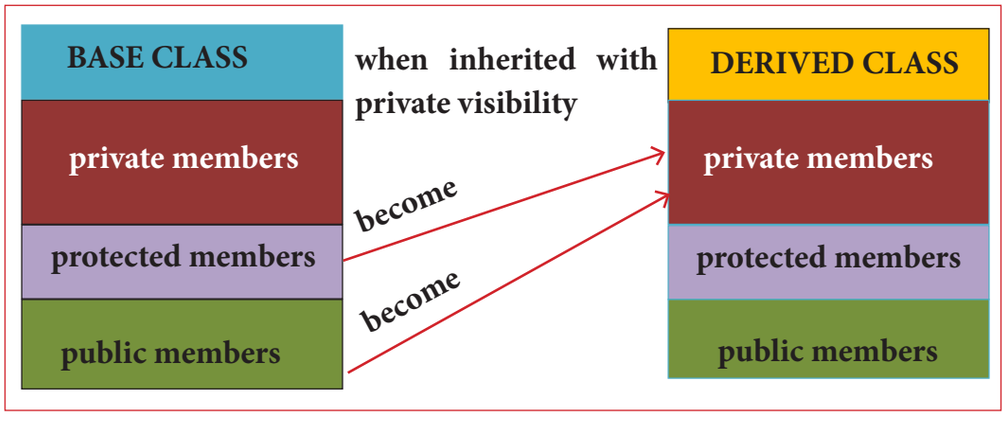
16.5 VISIBILITY MODES

An important feature of Inheritance is to know which member of the base class will be acquired by the derived class. This is done by using visibility modes.

The accessibility of base class by the derived class is controlled by visibility modes. The three visibility modes are private, protected and public. The default visibility mode is private. Though visibility modes and access specifiers look similar, the main difference between them is Access specifiers control the accessibility of the members with in the class where as visibility modes control the access of inherited members with in the class.

16.5.1 Private visibility mode

When a base class is inherited with **private** visibility mode the public and protected members of the base class become **'private'** members of the derived class



Chapter 16 Page 260-274.indd 265 3/24/2020 9:23:56 AM